

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 5 · Correlation and basic regression

TUESDAY SEP 22 AND THURSDAY SEP 24, 2026

WHAT YOU WILL BE ABLE TO DO BY MONDAY NIGHT

- 1** Fit a line to two columns of data. Say in your own words what the slope, intercept, and residual mean.
- 2** Read a regression output table and explain what standard error, the confidence interval, and the p value tell you.
- 3** Understand why coefficients can clear 0.05 and still not be worth acting on due to chance, lurking variables, and small effect sizes.



UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials.



3 Confirm that weeks/week05/ now exists in your repo.

It holds this week's README, the worksheet you fill in on Thursday, and the two Case Study data files.

01

CORRELATION AND THE REGRESSION LINE



A TEAM IS 50 AND 31 AFTER 81 GAMES. HOW MANY OF ITS LAST 81 DO YOU EXPECT IT TO WIN?

PREDICTION

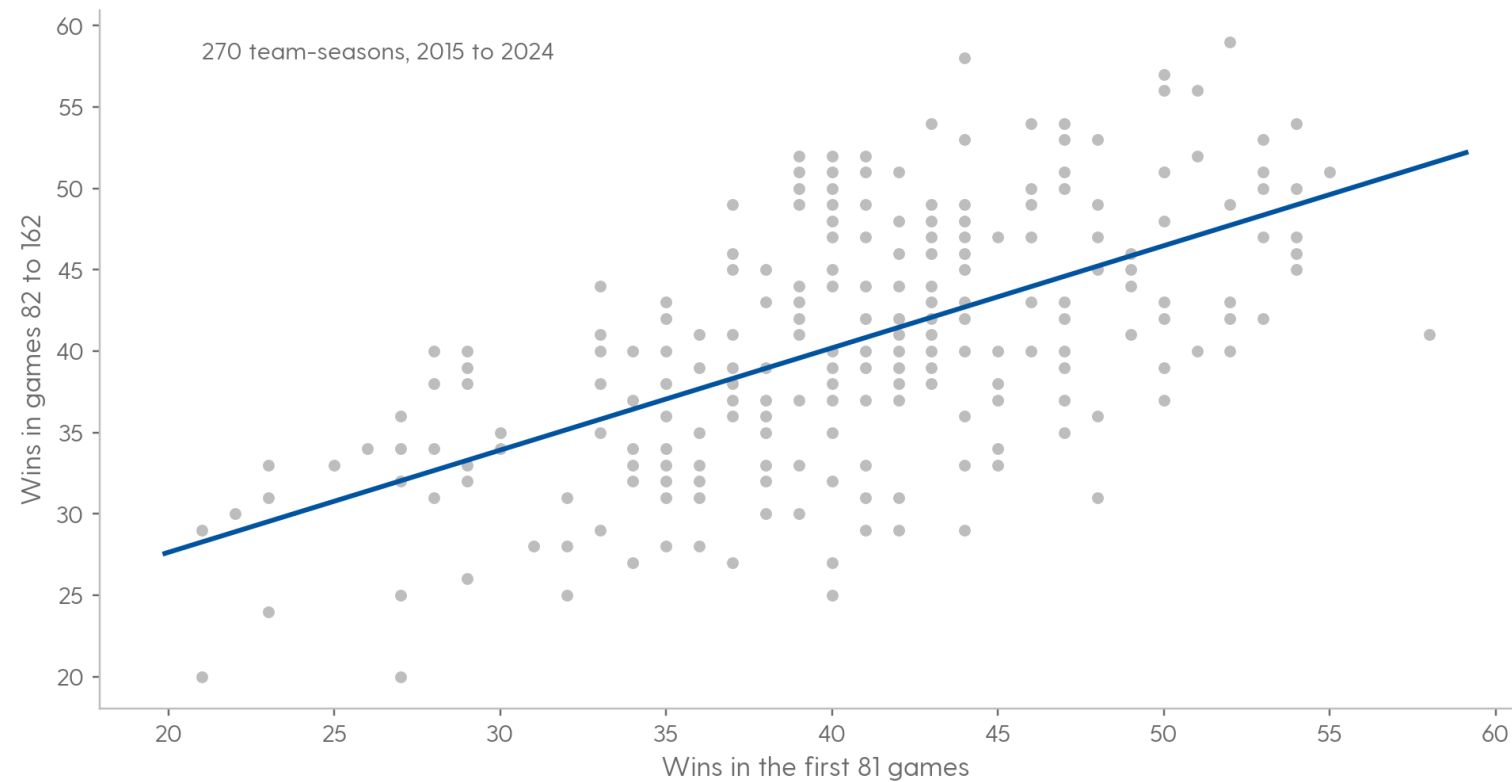
2024 TEAMS, BY HALF

TEAM	WINS IN THE FIRST 81	WINS IN GAMES 82 TO 162
St. Louis Cardinals	42	41
San Diego Padres	41	52
Tampa Bay Rays	40	40
Houston Astros	40	48

All four split the first half within 2 games of each other.

CORRELATION

FIRST-HALF WINS AND SECOND-HALF WINS, 2015 TO 2024



270 team-seasons. The correlation between the two halves is $r = 0.58$.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

CORRELATION (R)

How tightly two columns move together, from -1 to 1. A correlation of 0.58 says the dots cluster around a rising line. It does not say how steep the line is.

$r = 1$ is a perfect straight line upward, $r = -1$ a perfect line downward, $r = 0$ no linear relationship at all. In Excel: CORREL(range1, range2).

THE REGRESSION LINE

$$y = a + b \times x$$

Our prediction of y is a y-intercept (a), plus a coefficient (b) multiplied by our independent variable (x).

prediction		intercept	coefficient	

$$\text{second-half wins} = 15.1 + 0.63 \times \text{first-half wins}$$

In Excel: SLOPE(y range, x range) gives b and INTERCEPT(y range, x range) gives a.

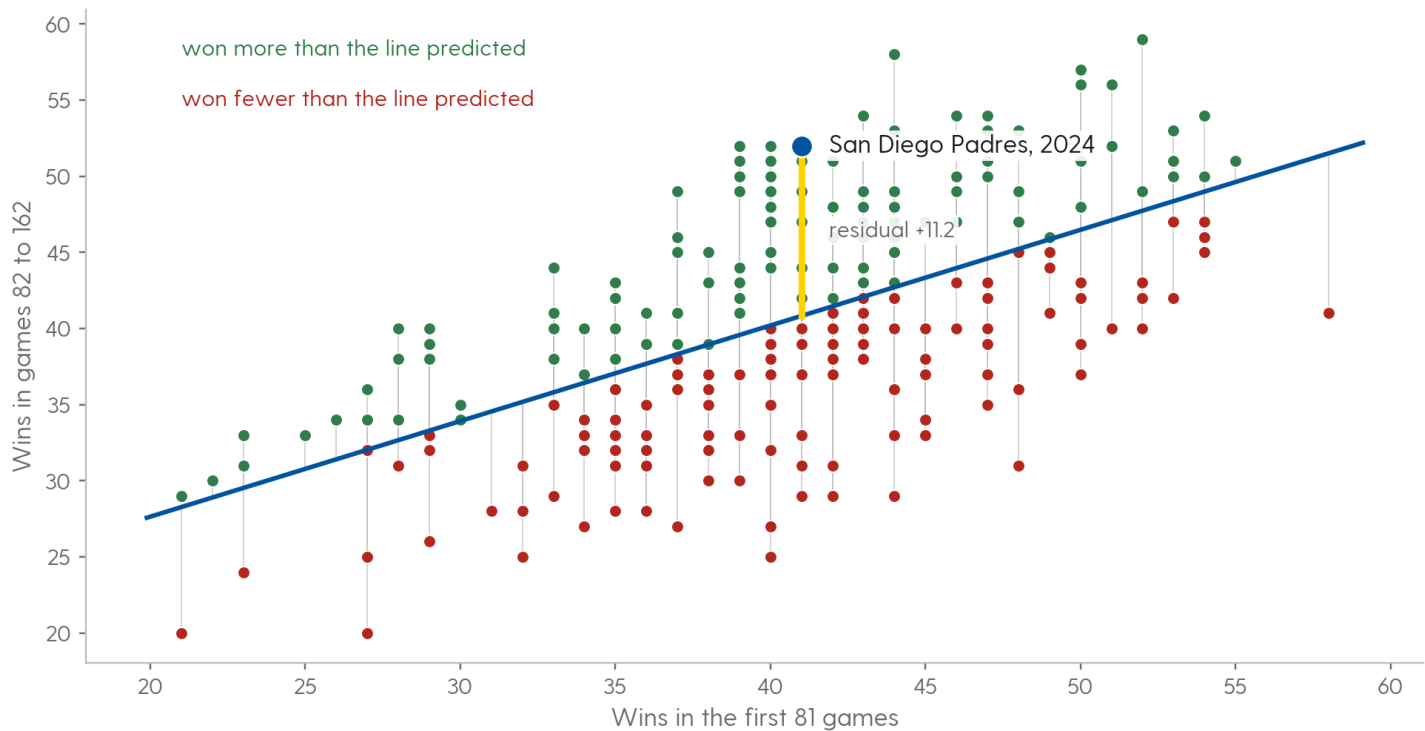


WHAT DOES THE 0.63 MEAN, IN WORDS?

For each increase in ____ (our independent variable), we expect an increase of ____ in ____ (our dependent variable).

RESIDUALS

SAN DIEGO PADRES, 2024



41 wins in the first half, so the line predicted 40.8 in the second. They won 52. The gap of +11.2 is the residual.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

RESIDUAL (ε)

Actual Y doesn't always equal $a + bx$. What's left is "random error" (ε). Sometimes this is due to randomness, other times it is due to factors outside of our model.

residual = actual y minus predicted y . The line is the one that makes the sum of the squared residuals as small as possible, which is why it is called least squares.

R SQUARED

34%

of the variation in second-half wins is accounted for by first-half wins. This is R squared, the correlation squared.

The other 66% is the residuals: injuries, trades, schedule, and luck. In Excel: `RSQ(y range, x range)`.



**WHAT ADDITIONAL FIRST-HALF TEAM DATA POINT COULD BETTER
PREDICT SECOND HALF TEAM PERFORMANCE?**

? IF WE PUT BOTH IN THE SAME MODEL, WHAT R SQUARED DO YOU EXPECT?

PREDICTORS DO NOT STACK.

Two columns that each explain a third of the variation do not explain two thirds together. They explain the same third. Adding a predictor is only worth it when it knows something the others do not.

02

READING THE REGRESSION OUTPUT

THE OUTPUT TABLE

SECOND-HALF WINS ON FIRST-HALF WINS, 270 TEAM-SEASONS

	INTERCEPT	FIRST-HALF WINS	WHAT IT MEANS
Coefficient	15.08	0.63	The slope: each first-half win is worth 0.63 second-half wins
Standard error	2.20	0.054	How much the coefficient would move in a new sample
Lower and upper 95%	10.75 to 19.41	0.52 to 0.73	The range the true coefficient falls in, with 95% confidence
t Stat	6.9	11.7	Standard errors the coefficient sits from zero
P-value	less than 0.0001	less than 0.0001	The chance of a slope at least this far from zero if there were no relationship between first- and second-half wins

This is the table Excel prints from Data Analysis, Regression, or from LINEST(y range, x range, TRUE, TRUE), with its columns down the side. MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

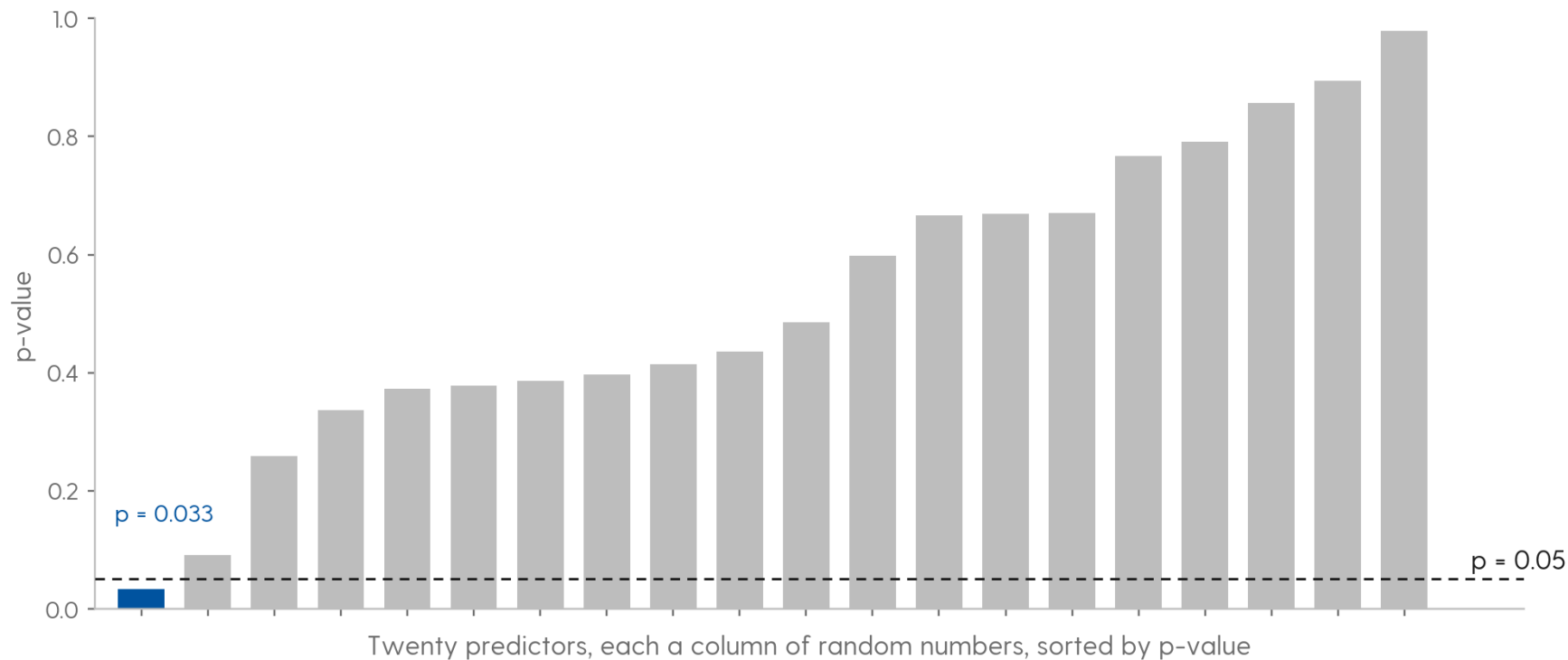
**THE NULL HYPOTHESIS: FIRST-HALF WINS TELL YOU
NOTHING ABOUT SECOND-HALF WINS.**

STATISTICALLY SIGNIFICANT

Typically, 0.05 is the P-Value used for "Statistical Significance". It means: if there were no relationship, a result this far from zero would show up less than one time in twenty.

If we tested twenty completely unrelated variables for correlation with Team Win % (Mascot Height, Letters in Team Name, etc.), we'd expect one of them to return as "statistically significant" by chance.

TWENTY RANDOM PREDICTORS OF 2024 WINS



Twenty columns of random numbers, each regressed on the 30 teams' win totals. 1 of the twenty crossed 0.05. Across 1,000 batches of twenty, at least one crossed in 64% of them.

Random predictors drawn with a fixed seed; wins from the MLB Stats API, 2024 regular season.

**STATISTICS DO NOT REPLACE CRITICAL THINKING OR
UNDERSTANDING THE SPORT!**

P-HACKING

Testing many things and reporting the one that cleared 0.05 as if it had been the only test. The chart before this one is what twenty tests look like when nothing is there.

Every extra predictor you try is another draw. Decide what you are testing before you look, and report how many things you tried.

BEYOND STATISTICAL SIGNIFICANCE

Spurious correlation

Mozzarella eaten per person and engineering bachelor's degrees awarded, $r = 0.97$. Ten years of national data, 2012 to 2021. Two series that both drift upward, and nothing behind it.

Lurking variable

Stork pairs and human birth rates across 17 European countries, $r = 0.62$, $p = 0.008$. It clears 0.05. Bigger countries hold more storks and more people, so country size moves both columns.

03

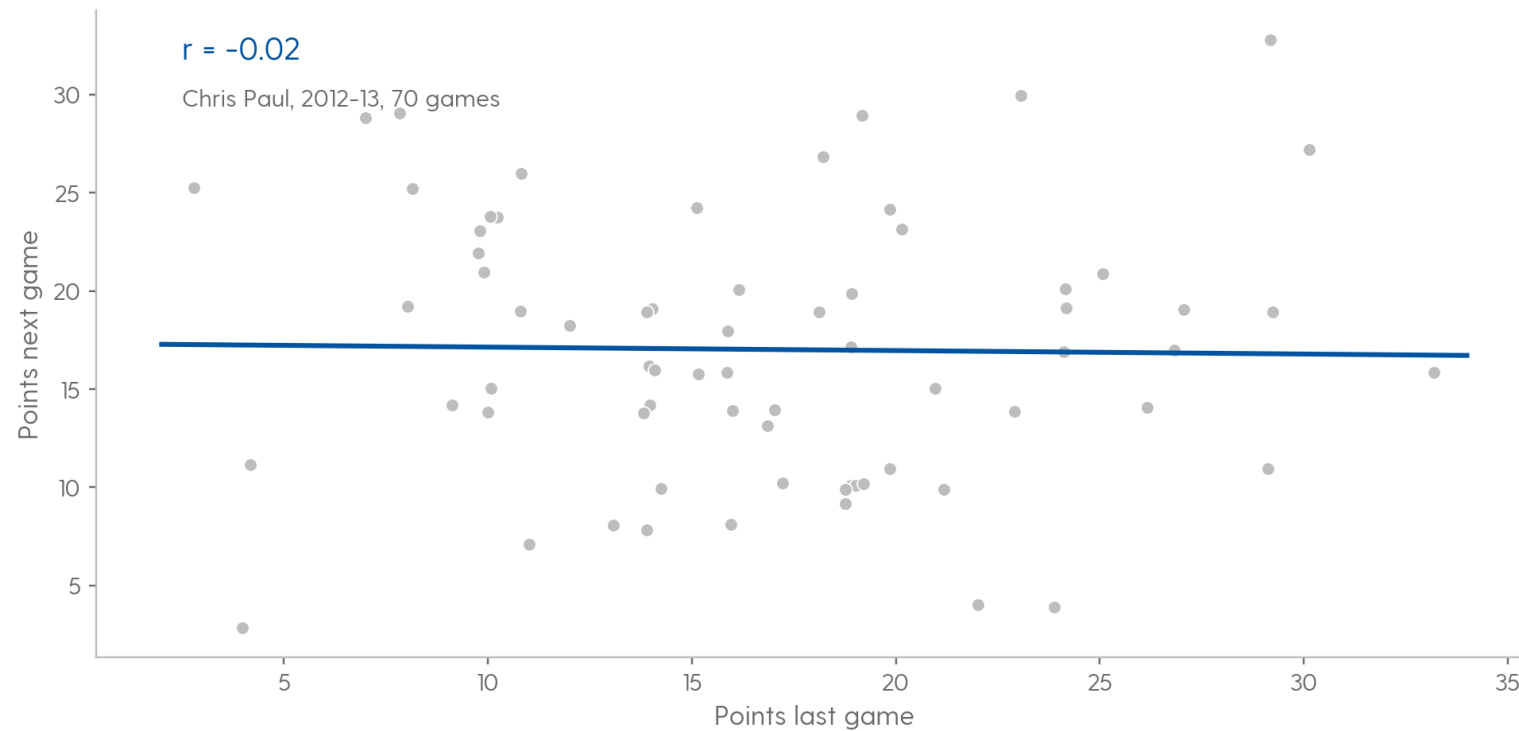
STATISTICAL AND PRACTICAL SIGNIFICANCE



AN NBA PLAYER SCORED MORE THAN HIS AVERAGE POINTS PER GAME IN A GAME LAST NIGHT. SHOULD I EXPECT HIM TO SCORE MORE IN OUR NEXT GAME?

AUTOCORRELATION

CHRIS PAUL, 2012-13



Each dot is one game's points against the next game's. 70 games, $r = -0.018$, $p = 0.89$. A big night tells you nothing about the next one.

Basketball-Reference. Game log, Los Angeles Clippers 2012-13.

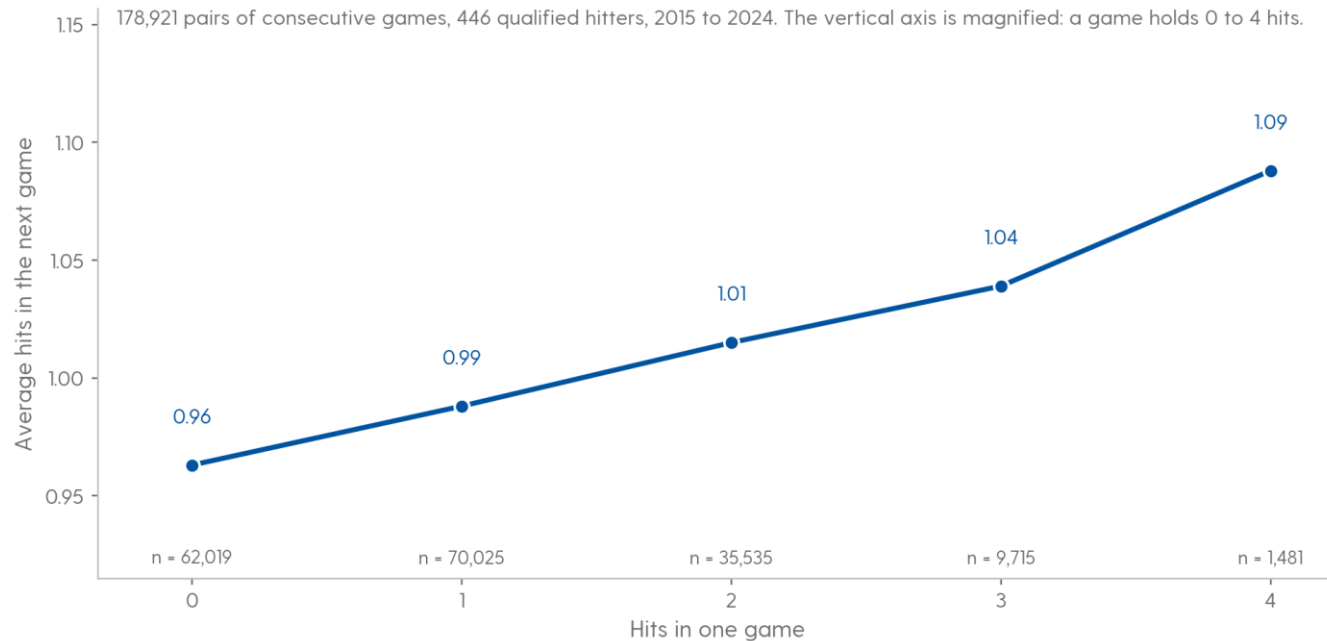
AUTOCORRELATION

The correlation of a series with itself one step later: this game against the next, this month against the next. A streaky player has a positive one.

In Excel: CORREL over two ranges offset by one row. In a model it is the residuals that matter: if this game's error is correlated with the last game's, the model is leaving predictable structure in the errors. The fix is a lagging variable, the previous game's value as a predictor, which moves that structure out of the error term and into the model.

WHO IS BATTING?

HITS IN ONE GAME AND HITS IN THE NEXT, 178,921 PAIRS



IF A PLAYER GETS 4 HITS IN A GAME, DO YOU THINK HE'S MORE LIKELY TO GET A HIT IN HIS NEXT GAME THAN HE WOULD'VE BEEN?

The average next game climbs 0.13 hits from a hitless game to a four-hit game. Across all 178,921 pairs the correlation is 0.027, the slope is 0.027 hits per hit, and the p-value is less than 0.0001.

MLB Stats API, game logs for every qualified hitter, regular seasons 2015 to 2024, 2020 excluded.

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WHO IS BATTING?

EACH HITTER AGAINST HIS OWN SEASON AVERAGE

PAIRS	R	P-VALUE	R SQUARED	SLOPE
178,921	0.005	0.044	0.002%	0.0048

It clears 0.05. The slope says a hitter who got three hits yesterday is expected to get 0.010 extra hits today, about one hundredth of a hit.

MLB Stats API, game logs for every qualified hitter, regular seasons 2015 to 2024, 2020 excluded. Each hitter's own season average is subtracted from both of his games before the correlation is taken.

STATISTICAL VS CLINICAL SIGNIFICANCE

With enough data, an effect that is nearly zero will still clear 0.05. Significance says the effect is probably not zero. It does not say the effect is big enough to act on, and it does not say you measured the right thing.

One hundredth of a hit is nothing a manager can put in a lineup, and the effect explains 0.002% of the variation. Read the coefficient and R squared first, then ask what else moves both columns.

04

CORRELATION AND CAUSATION

? NAME A STATISTIC THAT GOES WITH WINNING WITHOUT CAUSING IT.

CORRELATION $\neq$ CAUSATION

QB Kneels and Wins

Attendance and Win %

Draft Position and Record

Good Vibes and Wins

These aren't happening by random chance, they are meaningfully related. They just aren't directly causal.

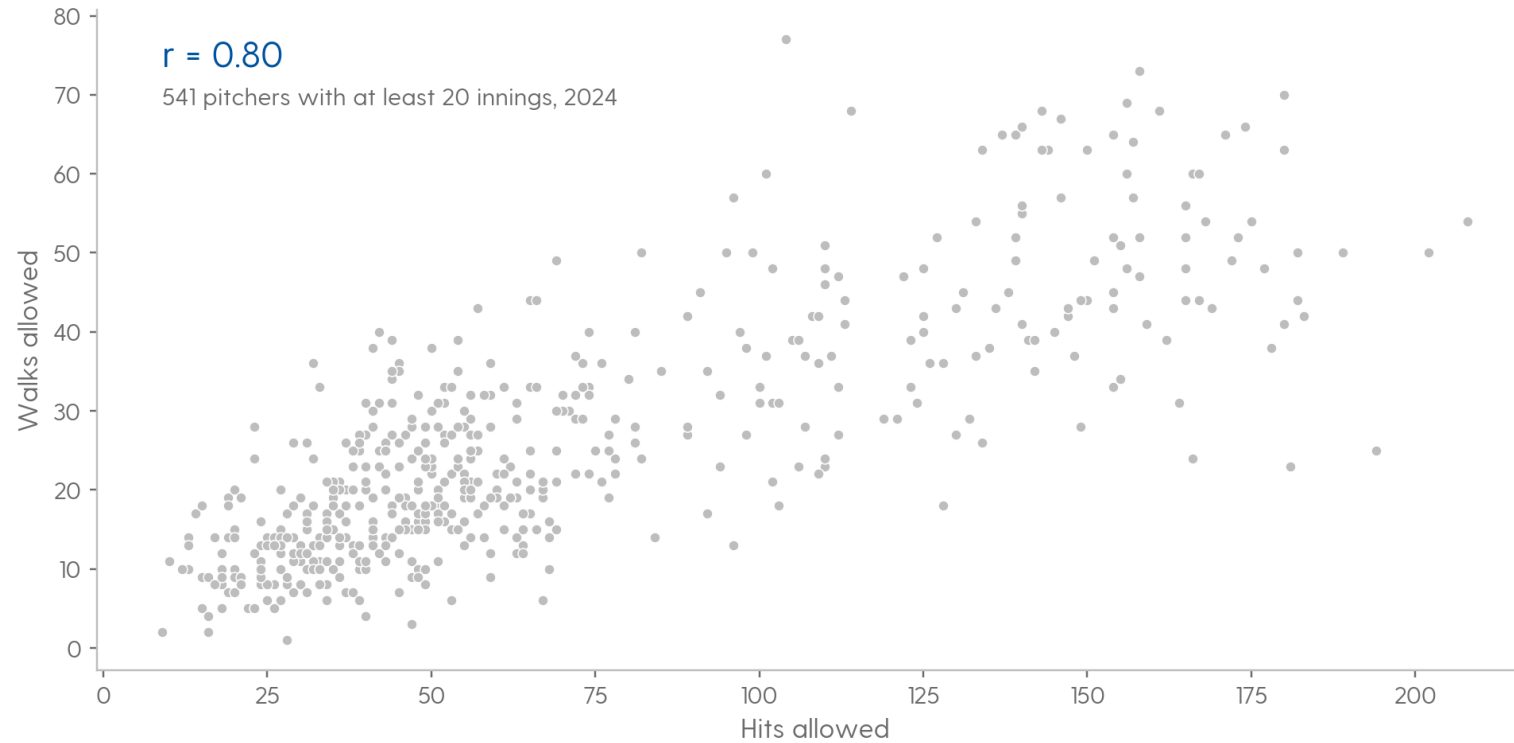
LURKING VARIABLE

A third thing that moves both of the two you measured.
Take it out and the relationship you saw can shrink or disappear.

Two ways to take it out: express both variables per unit of the third, or put the third variable in the regression as another predictor.

? ARE WALKS ALLOWED AND HITS ALLOWED CORRELATED ACROSS MLB PITCHERS?

HITS AND WALKS ALLOWED, 541 PITCHERS WITH AT LEAST 20 INNINGS



The two counts correlate at $r = 0.80$. Pitchers who give up hits also walk people.

MLB Stats API, 2024 regular season.

PARTIAL CORRELATION: HITS AND WALKS WITH INNINGS TAKEN OUT

$$r(x, y \mid z) = (r(x, y) - r(x, z) \times r(y, z)) \div \sqrt{((1 - r(x, z)^2)(1 - r(y, z)^2))}$$

x and y are the two things you measured. z is the third thing you suspect moves both of them.

$$r(H, W \mid I) = (r(H, W) - r(H, I) \times r(W, I)) \div \sqrt{((1 - r(H, I)^2)(1 - r(W, I)^2))}$$

$$= (0.80 - 0.96 \times 0.83) \div \sqrt{((1 - 0.96^2)(1 - 0.83^2))}$$

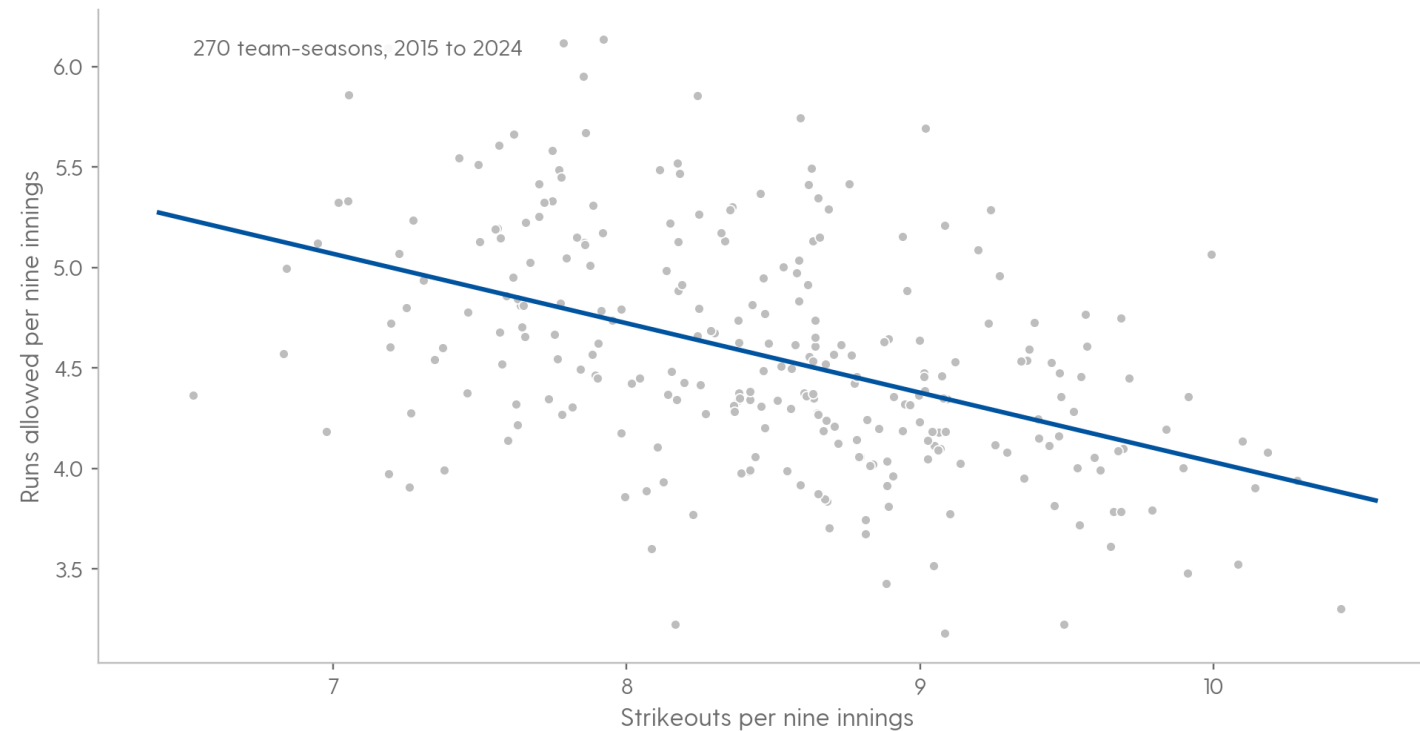
The correlation between hits and walks once innings pitched is held fixed. Same answer as the per-nine chart, by a different route.

$$R(H, W \mid I) = 0.02$$



DO STRIKEOUT PITCHERS ALLOW FEWER RUNS? WHAT ELSE WOULD YOU WANT IN THAT MODEL?

TEAM RUNS ALLOWED AND STRIKEOUTS PER NINE, 270 TEAM-SEASONS

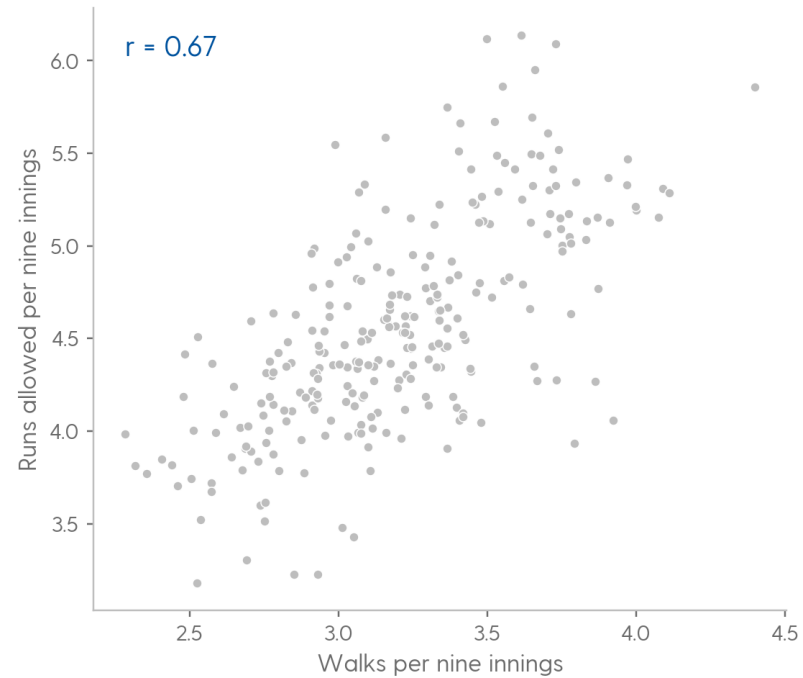
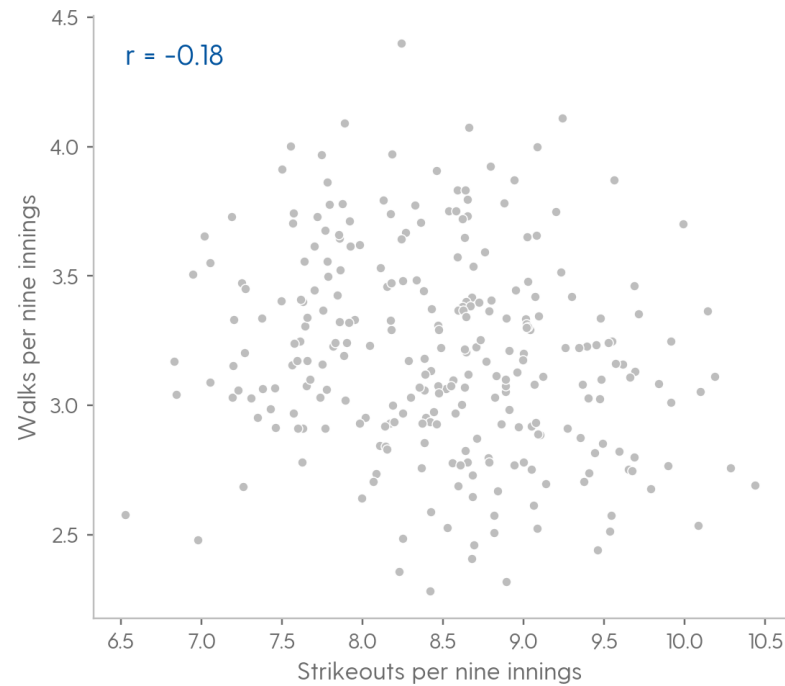


They correlate at $r = -0.46$, and the line says each strikeout per nine is worth -0.35 runs. On its own the model explains 21% of the variation.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

LURKING VARIABLES

WALKS PER NINE, AGAINST STRIKEOUTS AND AGAINST RUNS



270 team-seasons, 2015 to 2024

Walks barely track strikeouts at $r = -0.18$, and they track runs allowed hard at $r = 0.67$. A team's walk rate is most of what the strikeout model was missing.

MLB Stats API, regular seasons 2015 to 2024, 2020 excluded.

05

RELATIVE IMPORTANCE



TEAM POINTS PER GAME ON EFFECTIVE SHOOTING PERCENTAGE AND SHOT ATTEMPTS. WHICH COEFFICIENT WILL BE BIGGER, AND WILL THAT MEAN IT MATTERS MORE?

NBA TEAM POINTS PER GAME, 2019-20 TO 2024-25

$$\text{points per game} = -73.2 + 182 \times \text{eFG\%} + 0.99 \times \text{FGA}$$

eFG% runs from 0 to 1, so the 182 is per whole unit: 1.8 points per percentage point.

FGA is attempts per game, so the 0.99 is 0.99 points per extra shot.

180 team-seasons, R squared 0.89. eFG% counts a three as one and a half makes, so it credits shot quality as well as accuracy. Both coefficients clear 0.05 easily, and the two numbers cannot be compared to each other: one is per unit of a percentage, the other per shot.

06

THIS WEEK'S WORK



WEEK 8 OF THE 2021 NFL SEASON

- 1 Open hand-build-regression-worksheet.xlsx and its first sheet, slope by hand.
- 2 Build the slope the long way from the deviations, then check it with SLOPE, INTERCEPT, CORREL and RSQ.
- 3 Predict the margin for a team that runs 30 times, and compute the residual for LAC.

The file is in weeks/week05/data/ after /update. Rush attempts are x and final margin is y. This is the same kind of line as the baseball one, on a season the Case Study does not use.



THE SAME TEN GAMES: STANDARD ERROR AND P

- 1 Open the second sheet, standard error and p.
- 2 Compute each residual, the residual standard error (check it with STEYX), and the standard error of the slope from DEVSQ.
- 3 Compute t, the p-value with T.DIST.2T and the 95% interval with T.INV.2T, then read the standard error out of LINEST.
- 4 Do the same for turnover differential, then multiply each slope by its predictor's standard deviation.

Turnover differential is takeaways minus giveaways, counted the way a box score counts them.



NEITHER SLOPE CLEARS 0.05, AND RUSHING LOOKS STRONGER PER STANDARD DEVIATION. WHICH IS MORE LIKELY TO ACTUALLY CHANGE THE MARGIN?



RUNS PER GAME ON OBP AND SLG, 30 MLB TEAMS, 2024

- 1 Open the third sheet, two predictors.
- 2 Check how closely OBP and SLG move together with CORREL, then fit each one alone with SLOPE and RSQ.
- 3 Fit both at once with LINEST and read the coefficients, standard errors and R squared out with INDEX.

LINEST(y range, both x columns, TRUE, TRUE). INDEX(..., 1, 2) is the OBP coefficient and INDEX(..., 1, 1) is the SLG coefficient: LINEST lists the x columns in reverse order.

**? EACH COEFFICIENT ROUGHLY HALVES WHEN THE OTHER IS ADDED.
WHY?**



CASE STUDY STEPS

- 1** Regress final margin on rush attempts across 2,174 team-games, 2022 to 2025, and read the output table.
- 2** Add the score the team was playing from, `mean_score_diff`, as a second predictor and compare the two tables.
- 3** Build an interactive scatter of the team-games with hover detail and a season control.



SECOND CASE STUDY STEPS

- 1** Predict wins in games 82 to 162 from first-half wins, then from first-half run differential. Those two are the bar.
- 2** Build something from the two MLB files that you think beats them, and check it with adjusted R squared rather than R squared.
- 3** Say what you tried, whether it worked, and what you would look at next.

The data is `mlb-team-games-2014-2024.csv` and `mlb-team-first-half-lines-2015-2024.csv`, both raw: every number you need is built from them, and none of it is a column you can read off.



DOES RUNNING THE BALL WIN GAMES, OR DOES WINNING MAKE YOU RUN?

- 1 When you considered the score, what happened to the rushing coefficient? Why?
- 2 Once you controlled for score, what was the rushing coefficient? Do you believe it is telling you something meaningful? Why?
- 3 What other data would you want to look at before reporting back to Coach Brady?

Written to Joe Brady, the new head coach of the Buffalo Bills, looking back at the seasons before him. python3 scripts/new_brief.py week05 writes BRIEF.md with all 6 questions as headings; /coach-brief 5 critiques a draft and will not write one.



BUY OR SELL AT THE DEADLINE?

- 4 What did you try adding to the model, and why did you think it would help?
- 5 Did it work? Say how you decided.
- 6 You are the GM deciding whether to buy or sell, and you are committed to getting this right. What would you look at next, and why that?

BEFORE YOU PUSH ON MONDAY, SEPTEMBER 28

Commit the filled-in worksheet in `weeks/week05/data/`.

Commit your analysis code and its outputs in `weeks/week05/outputs/`.

Answer every question in `BRIEF.md`, in your own words.

Run `/audit`, then `/quiz-me`.

Upload your repository's link to Canvas by Monday, September 28 at 11:59pm.